

Network and Systems Administrator Capstone Project

Final Project

Infrastructure Design for Vantage IT Solutions

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TABLE OF CONTENTS

- I. Organization and Requirements
- II. Equipment Inventory
- III. Final Physical and Logical Topologies
- IV. Headquarters and Branch Design
- V. WAN and VPN
- VI. VLAN and IP-Addressing Tables
- VII. Device and Port Assignments
- VIII. Virtualization
- IX. Services and Services
- X. Security
- XI. Backup and Recovery
- XII. Monitoring and Logging
- XIII. Testing and Troubleshooting
- XIV. Risk Register
- XV. Individual Work Logs
- XVI. References

OVERVIEW

This document reports the complete IT infrastructure solutions for Vantage IT Solutions a small fictional organization. This project is written for the purpose of fulfilling the final practical assessment for the Network and Systems Administrator Program.

1. Organization and Requirements

1.1 Company Overview

Vantage IT solutions is a small fictional start up IT solution company. The main objective of the company is to provide practical solutions, small office network deployments, network administration, and IT consulting for local small businesses companies. There are two locations that the company use, a headquarters office for administration and centralized IT services, and also a small branch location that provides local support for the clients. The company comprise of less than 30 staffs for both headquarters and branch.

1.2 Department and User-groups

Department	Users	Site	Access Need
Management	2	HQ	Sensitive management document
Sales	4	HQ/Branch	Clients' relationship management
Projects	5	HQ	Network and system deployment, project files
Service & Support	5	HQ/Branch	Remote Assistance, ticketing, remote support
Finance/HR/Administration	3	HQ	Administrative documents, HR records, payroll
Branch Coordination	1	Branch	Branch operation and local operations
IT administration	2	HQ	Infrastructure Administration

1.3 Organizations Structure

Level	Roles
Executive	General Manager
Business	Sales & Client Services; Admin/Finance/HR
Operations	Project lead; Service support lead; Branch Coordinator
Technology	IT administrator & assistant
Users	Staff members

Management Department

Albert.Sterling@vantageit.local

Olivia.Vance@vantageit.local

Sales Department

Kathy.Brooks@vantageit.local

Mark.Dawson@vantageit.local

Taylor.Caldwell@vantageit.local

Albert.Kensington@vantageit.local

Projects Department

Noah.Barrett@vantageit.local

Jordan.Hayes@vantageit.local

Amelia.Croft@vantageit.local

Ava.Mercer@vantageit.local

Liam.Dalton@vantageit.local

Service and Support Department

Riley.Harper@vantageit.local

Alex.Stanton@vantageit.local

Casey.Winslow@vantageit.local

Peyton.Blair@vantageit.local

Logan.Rhodes@vantageit.local

Finance, HR, and Administration Department

Miley.Carmichael@vantageit.local

Hillary.Preston@vantageit.local

Donald.Russo@vantageit.local

IT Administration Department

evan.warkentin@vantageit.local

neil.montances@vantageit.local

1.4 Requirements

- All staff and personnel must be authenticated on one centralized server
- Authorized staff for both sites may access shared files, internal email and web application, and the DNS services
- Branch users must be securely accessing approved headquarters services
- IT personnel have a reliable inventory system, resource monitoring, a secure remote administration, and a log system
- Data configurations and project documentation must have a repeatable backup and recovery procedure

1.5 Technical Requirements

Area	Decision / Requirement
Identity	Microsoft ADDS at HQ; standard/admin accounts; workstations domain joined
DNS	AD-integrated internal DNS on HQ-DC01; branch clients query HQ DNS across VPN
DHCP	Central Microsoft DHCP; HQ FortiSwitch relay and branch Cisco IP helper-address
Routing	HQ L3 Switch provides HQ SVI gateways/inter-VLAN routing; branch Cisco router uses 802.1Q sub-interfaces
WAN	Use of the /30 links only for external transit
VPN	IKE/IPsec site-to-site between HQ-Router and BR-Router; services restrictions for 10.10.0.0/16 and 10.20.0.0/16
Virtualization	Proxmox, 1 dedicated port connected to HQ-L2-SW01
File/NTP	Ubuntu Samba AD member and Chrony server on HQ-FILE01
Mail	Exchange Server 2019, internal email
Web	NGINX reverse proxy to a small internal service portal
Monitoring	LibreNMS with SNMPv3 preferred and for system logs
Inventory	Snipe-IT
Security	Management isolation, secure protocols, disabled all unused ports, backups, endpoint/server firewalls, authorized vulnerability assessment

1.6 User Roles

Account / Role	Purpose	Key restrictions
Standard domain user	Email, file access, intranet, normal workstation tasks	No local/server/network-device administrative rights
Department security group	Controls access to departmental resources	Membership follows job role and manager approval
IT admin account	Infrastructure/server administration	Used only for administrative tasks; MFA where supported; not used for email/web browsing
Network admin account	Cisco/Fortinet/Proxmox device management	Allowed only from management VLAN or approved admin workstation
Service account	Application or scheduled service identity	No interactive logon unless specifically required; least privilege
Backup operator	Backup media rotation and restore tests	Access limited to backup workflow and protected media

1.7 Naming Conventions

Object	Standard	Examples
Sites	HQ, BR	HQ-Router, BR-Router
Network devices	<SITE>-<ROLE>-<NN>	HQ-L3-SW01, HQ-L2-SW02, BR-L2-SW01
Servers	HQ-<SERVICE>-<NN>	HQ-DC01, HQ-FILE01, HQ-NMS01
Workstations	<SITE>-WS<NN>	HQ-WS01, BR-WS01
Access points	<SITE>-AP	HQ-AP, BR-AP
Users	first.last	alex.lee
Admin users	adm-first.last	evan.admin (sample format)
Service accounts	svc-<service>	svc-backup
AD groups	GG-<purpose>-<access>	GG-File-Projects-RW
Domain	vantageit.local	Internal AD/DNS namespace
Email	first.last@vantageit.local	Internal-only mail domain

2. Equipment Inventory

Asset Tag	Item	Type	Qty	Site	Hostname / Label	Usage
VIT-NET-001	Cisco ISR 4321	Router	1	HQ	HQ-Router	WAN edge and VPN endpoint
VIT-NET-002	Cisco 2900 Series	Router	1	Branch	BR-Router	WAN edge, branch VLAN gateways, VPN endpoint
VIT-NET-003	FortiSwitch 124F-FPOE	Layer 3 switch	1	HQ	HQ-L3-SW01	HQ core, SVI gateways, inter-VLAN routing
VIT-NET-004	Cisco Catalyst 2960	Layer 2 Switch	1	HQ	HQ-L2-SW01	Proxmox + wired access
VIT-NET-005	Cisco Catalyst 2960	Layer 2 Switch	1	HQ	HQ-L2-SW02	HQ users + access point
VIT-NET-006	Cisco Catalyst 2960	Layer 2 Switch	1	Branch	BR-L2-SW01	Branch wired/AP access
VIT-WLAN-001	Linksys EA6300	Wireless Access Point	1	HQ	HQ-AP	Wireless access only
VIT-WLAN-002	TP-Link Archer AX23	Wireless Access Point	1	Branch	BR-AP	Wireless access only
VIT-CMP-001	HP ProBook, 32 GB RAM, 1 TB SSD, Core Ultra 5	Laptop/virtualization host	1	HQ	HQ-PVE01	Proxmox host (VE)
VIT-CMP-002	HP ProBook	Laptop/workstation	1	HQ	HQ-WS01	HQ domain workstation
VIT-CMP-003	HP ProBook	Laptop/workstation	1	Branch	BR-WS01	Branch domain workstation
VIT-MED-001/2	USB flash drive	Backup media	2	HQ	Backup-A / Backup-B	Encrypted offline backup rotation
VIT-CAB-(001 to 014)	Cat6 patch cable	Cables	14	Both	Patch cables	Device-to-patch-panel / rack patching

- Assets records are log and maintained through Snipe-IT

3. Final Physical and Logical Topologies

4. Headquarters and Branch Design

5. WAN and VPN

6. VLAN and IP-Addressing Table

6.1 Headquarters

VLAN	Name	Subnet	Gateway	Addressing	Purpose
10	MGMT	10.10.10.0/24	10.10.10.1	Static	Management for Network Devices
20	SERVERS	10.10.20.0/24	10.10.20.1	Static	Centralized Servers/VMs
30	STAFF	10.10.30.0/24	10.10.30.1	DHCP	Wired Users/Printers
35	VIT-WIFI	10.10.35.0/24	10.10.35.1	DHCP	HQ-User Wifi access
40	IoT-CCTV	10.10.40.0/24	10.10.40.1	DHCP	Future recommendation
50	GUEST-WIFI	10.10.50.0/24	10.10.50.1	DHCP	Reserved guest subnet; Internet only when VLAN-capable AP exists
99	NATIVE-UNUSED	No client subnet	None	None	Native VLANs / Unused ports

6.2 Branch VLANs

VLAN	Name	Subnet	Gateway	Addressing	Purpose
110	MGMT	10.20.10.0/24	10.20.10.1	Static	Branch network management
130	WIRED-STAFF	10.20.30.0/24	10.20.30.1	DHCP via relay	Branch wired users/printer
135	CORP-WIFI	10.20.35.0/24	10.20.35.1	DHCP via relay	Branch corporate wireless
150	GUEST-WIFI	10.20.50.0/24	10.20.50.1	DHCP via relay/local alternative	Internet-only guest network if hardware supports
199	NATIVE-UNUSED	None	None	None	Native/Unused

6.3 DHCP Scope

Scope	Range	Router	DNS	Lease	Notes
HQ-STAFF-V30	10.10.30.50-220	10.10.30.1	10.10.20.10	8 days	.1-.49 reserved .221-.254 spare
HQ-WIFI-V35	10.10.35.50-220	10.10.35.1	10.10.20.10	3 days	HQ wireless
HQ-IOT-V40	10.10.40.50-200	10.10.40.1	10.10.20.10	8 days	For future use
HQ-GUEST-V50	10.10.50.50-230	10.10.50.1	10.10.20.10	8 hours	Internet-only
BR-WIRED-V130	10.20.30.50-220	10.20.30.1	10.10.20.10	8 days	Relayed across VPN
BR-WIFI-V135	10.20.35.50-220	10.20.35.1	10.10.20.10	3 days	Relayed across VPN
BR-GUEST-V150	10.20.50.50-230	10.20.50.1		8 hours	Prefer local fallback DHCP if guest must survive VPN loss

7. Device and Port Assignment

8. Virtualization

9. Servers and Services

10. Security

11. Backup and Recovery

12. Monitoring or Logging

13. Testing and Troubleshooting

14. Risk Register

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16. References